

Notes on Cacciatore and Candian (2025)

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Motivation In 2023, supply shortages and rising prices in key upstream industries became primary drivers of the global inflation surge.

Research Question How does uncertainty arising in a particular industry affect aggregate uncertainty, production, and prices in upstream suppliers and downstream customers? Do aggregate effects of uncertainty depend on where in the production network it originates? ¹

Empirical Results The volatility measures (implied volatility, realized volatility) for firm f , industry i , date d , period t are

$$IV_{i,d} = \sum_{f \in i} s_{f,i,d-1} IV_{f,i,d}, \quad RV_{i,t} = \frac{1}{D} \sum_{d \in t} \sum_{f \in i} s_{f,i,d} r_{f,i,d}^2,$$

where r is the stock return.

- (a) Sectoral volatility component explains a large part of firm implied volatility. Different sectors may have different spikes.
- (b) Cyclicalities: Negative correlation in general with industrial production; positive correlation with aggregate uncertainty; no clear correlation with core CPI inflation.

¹This uncertainty is literally the implied volatility from the stock market, not the volatility of productivity.

- (c) Uncertainty shock: Persistently higher uncertainty in both upstream and downstream industries and own industry. Also lower employment. Lower the producer prices in upstream but raise in own industry and downstream. Intermediate input prices (index) drops in upstream and raises in downstream.
- (d) Aggregate effect: Uncertainty rises, employment drops, and CPI increases.

Theories

- (a) There is a wholesaler and retailers which I think are kind of redundant?
- (b) Productivity processes are standard.
- (c) Government purchase is formally modelled with an underlying process. But I prefer the fiscal policy as tax and rebated to household as in (La'O and Tahbaz-Salehi, 2022).

References

- Cacciatore, Matteo and Giacomo Candian**, “Uncertainty Through the Production Network: Sectoral Origins and Macroeconomic Implications,” *NBER Working Paper Series*, June 2025, 33953.
- La'O, Jennifer and Alireza Tahbaz-Salehi**, “Optimal Monetary Policy in Production Networks,” *Econometrica*, 2022, 90 (3), 1295–1336.